

Hugging Face Model Research and Acquisition Guide

Fresh provider verification and HF-managed deployment

ARCHITECTURE

HF-MANAGED

EVIDENCE-FIRST

PRODUCTION GATES

Authoritative decision. VeriTrust executes no learned production inference in browser code, Vercel functions, platform workers, local runtimes, or VeriTrust-owned model servers. Learned production inference uses Hugging Face-managed infrastructure. The canonical hierarchy is HF-operated hf-inference when the exact model/task and all gates pass; then Hugging Face Inference Endpoint; then private Hugging Face repository plus Hugging Face Inference Endpoint. Third-party Hugging Face Inference Providers are DISABLED_BY_DEFAULT, outside the authoritative production path, and never an automatic fallback.

Status: implementation architecture and research package; no production performance claim. All UNKNOWN values remain explicit.

Prepared from the executable repository, seven historical PDFs, fresh official Hugging Face research, primary competitor sources and C2PA standards.

Contents and control

This is an Implementation Architecture Handbook, not a standalone executable specification. Critical invariants, model identities, deployment classes, evidence semantics, receipt fields, lifecycle states and companion references are generated from validated canonical structured sources; all provider availability is point-in-time and promotion-gated.

NORMATIVE IMPLEMENTATION COMPANIONS: `Architecture_Source_of_Truth.yaml`; `Model_Registry_Proposal.json`; `HuggingFace_Deployment_Matrix.csv`; `Implementation_Backlog.csv`; `Database_Target_Contract.md`; `Verification_Test_Plans.md`; `Migration_Roadmap.md`.

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Cross-document invariants

- HF-managed learned inference only; no normal third-party provider tier.
- Registry resolution precedes dispatch.
- Receipt identifies exact deployment, bundle and ordered learned components.
- `raw_output`, `raw_label`, `raw_score`, `normalized_label`, `normalized_score`, `calibrated_probability` and `calibration_version` remain distinct.
- Expected-evidence lifecycle uses the canonical 16-state enum: `planned`, `dispatched`, `running`, `completed`, `abstained`, `not_applicable`, `unavailable`, `timed_out`, `failed`, `cancelled`, `privacy_blocked`, `policy_skipped`, `insufficient`, `unsupported`, `degraded`, `budget_exceeded`.
- Specialist evidence cannot silently substitute or own final policy.
- No original PDF or production source file is overwritten.

1. Research method and authoritative deployment policy

Research first, deploy second. Provider mapping is point-in-time and must be rechecked before promotion.

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On 16 August 2026, official Hub APIs were queried with ``expand=inferenceProviderMapping``; default metadata, configs, processors/tokenizers and model cards were saved under Research and pinned to immutable commits. Absence of a mapping is recorded as no verified serverless support-not proof that deployment is impossible.

2. HF deployment hierarchy

Third-party Inference Providers are **DISABLED_BY_DEFAULT, outside the authoritative production path and never an automatic fallback. Future exception only after an explicit product-owner architecture decision plus security, privacy and legal approval.**

Order	Mode	Use
1	HF-operated hf-inference when the exact model/task and all gates pass	Exact model and task have a fresh, explicit hf-inference mapping; golden contract, privacy, latency, quota, and cost gates pass.
2	Hugging Face Inference Endpoint	Default for public models without verified serverless support; commit revision, endpoint deployment, and response schema are pinned.
3	private Hugging Face repository plus Hugging Face Inference Endpoint	For VeriTrust fine-tunes, private learned heads, GBDT/ONNX models, or controlled copies with custom handlers.

3. Official HF platform findings

- HF Inference is the serverless service and currently focuses mostly on CPU-compatible tasks.
- Inference Endpoints support commit revisions, provider/region/instance selection, autoscaling and protected/private modes.
- Scale-to-zero saves idle cost but introduces cold starts and may return 502 while replicas initialize.
- Custom ``handler.py``/requirements and custom containers support preprocessing, nonstandard tasks and private GBDT/ONNX.
- Endpoint/provider logs may exist for up to 30 days per public docs; provider-specific terms and DPA still require review.
- OpenMetrics/analytics expose request, latency and hardware data; VeriTrust still owns evidence/audit telemetry.

4. Fresh provider verification

Repository	Revision	Mapping	Decision
ealvaradob/bert-finetuned-phishing	fa8fb73	hf-inference	serverless candidate
cybersectony/...v2.4.1	6724dec	hf-inference	shadow; labels ambiguous
CrabInHoney/urlbert...v4	917ded0	none	HF Endpoint
yermandy/deepfake-detection	9a6857e	none	private custom Endpoint
OwensLab/commfor-model-384	6076002	none	custom Endpoint
mikaelnurminen/modernbert...	d28c16f	none	shadow Endpoint

5. Deepfake dossiers

`yermandy/deepfake-detection` is MIT and uses custom CLIP/PEFT behavior; the Hub files include unsafe/suspicious serialized formats, so a controlled private copy and custom-handler review are mandatory. `OwensLab/commfor-model-384` and `-224` are MIT safetensors but use custom model/preprocessor behavior. Community Forensics dataset licensing is per-source.

- Face manipulation and full-frame synthesis are not substitutes.
- No card metric is a VeriTrust benchmark.
- A SigLIP2 base checkpoint needs a trained/private OOD head before it is an OOD detector.

6. Phishing dossiers

`Sonje03/phishlens-distilbert` is an email-specific English DistilBERT candidate with no current mapping and therefore targets an HF Endpoint. `ealvaradob/bert-finetuned-phishing` is a larger broad-modality English BERT candidate with an HF-operated hf-inference mapping. Both remain promotion-gated; model selection precedes deployment selection. `cybersectony/...v2.4.1` remains shadow because label semantics need proof; ModernBERT remains a semantic shadow.

- No candidate disappears because it lacks serverless support.
- microsoft/mdebarta-v3-base is a base checkpoint, not a phishing classifier.
- All candidate licenses/datasets/derivatives require recorded legal review.

7. Complete investigated-model decisions

No scientifically serious repository may disappear silently from the acquisition record.

The authoritative disposition for all 15 preserved Hugging Face repositories is loaded from HF_Model_Decision_Ledger.csv. Every repository under Research/Sources/HuggingFace/models has an explicit decision and reason.

Repository / immutable revision	Decision	Reason / deployment implication
CrabInHoney/urlbert-tiny-v4-malicious-url-classifier @ 917ded0543a630d6e82570bb01ae692f6cbb95f1	SELECTED_PRIMARY_CAN DIDATE	role-specific compact classifier with exact label map but no VeriTrust benchmark; protected HF Inference Endpoint after qualification

Repository / immutable revision	Decision	Reason / deployment implication
cybersectony/phishing-email-detection-distilbert_v2.4.1 @ 6724dec17360ab1f6450553f1d950404f66a5dde	SELECTED_SHADOW	current mapping but generic/ambiguous four-label semantics block authority; HF-operated hf-inference shadow only after golden labels
dima806/phishing-email-detection @ 310a5602280486e01dd60e37230980cfcc327b41	RESEARCH_ONLY	retained for comparative benchmarking but weaker preserved provenance than promotion-gated fast pair; HF Endpoint only if research reopens it
ealvaradob/bert-finetuned-phishing @ fa8fb73a007174c410ab7160d4e4c6e6b8d998d4	UNRESOLVED	promotion-gated head-to-head with PhishLens; mapping does not decide scientific selection; HF-operated hf-inference only if selected and all gates pass
google/siglip2-base-patch16-224 @ 75de2d55ec2d0b4efc50b3e9ad70dba96a7b2fa2	SELECTED_EMBEDDING_ROLE_AND_BASE_FOR_FINE_TUNE	selected for supporting Link similarity and separate private OOD research with distinct keys/contracts; HF Endpoint for Link embedding; private HF repo plus Endpoint for trained OOD head
microsoft/deberta-v3-small @ a36c739020e01763fe789b4b85e2df55d6180012	SELECTED_BASE_FOR_FINE_TUNE	base checkpoint retained for a separately trained private role; private HF repo plus HF Endpoint after training
microsoft/mdeberta-v3-base @ a0484667b22365f84929a935b5e50a51f71f159d	SELECTED_BASE_FOR_FINE_TUNE	base checkpoint retained for separately trained multilingual phishing role; private HF repo plus HF Endpoint after training
mikaelnurminen/modernbert-large-phishing @ d28c16f882454115d71375d3321fbfd90e566490	SELECTED_SHADOW	retained as promotion-gated semantic shadow for error analysis; protected HF Inference Endpoint shadow
onnx-community/CommunityForensics-DeepfakeDet-ViT-ONNX @ f9fc6516b37cd7d1bec3b94847b3682ce97249f0	SUPERSEDED	official OwensLab checkpoints are preferred acquisition sources; conversion adds no independent family; do not deploy in normal portfolio; retain research record
OwensLab/commfor-model-224 @ 26afc31e6b40c312c3fd42c05a758be62446215b	SELECTED_SHADOW	latency/synthesis shadow only and explicitly not independent forensics; protected custom HF Inference Endpoint shadow
OwensLab/commfor-model-384 @ 6076002bf0d9dd37537f965ee2f06f826c333b61	SELECTED_PRIMARY_CANDIDATE	retained as promotion-gated full-frame synthesis candidate; protected custom HF Inference Endpoint
pirocheto/phishing-url-detection @ 44f3b19f705b52532e0aadf3d0d15dd892b8a2fb	RESEARCH_ONLY	unsafe pickle is excluded and ONNX contract requires separate security/scientific proof; custom HF Inference Endpoint only for research
prithivMLmods/Al-vs-Deepfake-vs-Real @ e1df5230c067ff4b189230343e548c365d80bbdb	RESEARCH_ONLY	retained for benchmark comparison but not independent low-level forensic evidence; HF Endpoint only if research reopens it
Sonje03/phishlens-distilbert @ b585a2b9bbf6777b344d017f4eaff9423363a489	UNRESOLVED	promotion-gated head-to-head with ealvaradob; absence of serverless mapping is not rejection; HF Inference Endpoint if selected
yermandy/deepfake-detection @ 9a6857ec642deb57373c5437be803a199468b8c6	SELECTED_PRIMARY_CANDIDATE	retained for the facial-manipulation role behind explicit HF-managed face preprocessing; controlled private HF repository plus private HF Endpoint

8. Link dossiers and exact label semantics

The current VeriTrust URLBERT mapping reverses defacement and phishing. This must be fixed before collecting qualification data.

`CrabInHoney/urlbert-tiny-v4-malicious-url-classifier` is Apache-2.0, tiny BERT, no provider mapping. Its canonical immutable label contract is: LABEL_0=benign; LABEL_1=defacement; LABEL_2=malware; LABEL_3=phishing. `pirocheto/phishing-url-detection` remains research-only on a custom HF Endpoint.

9. Private model workflow

Each private model requires dataset manifest, training code revision, feature schema, base revision, seed/reproducibility, artifact hash, evaluation/calibration, model card, security scan, private HF repository, immutable revision, HF endpoint, golden tests, shadow, canary, promotion and rollback.

Private role	Base/artifact	HF deployment
Multilingual phishing	mDeBERTa base + trained head	private repo + endpoint
Deepfake OOD	SigLIP2 + trained OOD head	private repo + endpoint
GBDT/ONNX	safe reviewed learned artifact	custom handler/container endpoint
Page text	DeBERTa base + trained classifier	private repo + endpoint

10. Request/response and registry contract

Endpoint adapters send versioned schemas and accept only bounded JSON/binary responses. The registry stores scientific role, repository/revision, endpoint deployment, preprocessing, label map, calibration and benchmark. Receipt mismatch fails closed.

Evidence	Fields
Required fields	evidence_id, investigation_id, artifact_id, specialist, scientific_role, registry_resolution_id, deployment_id, model_key, immutable_model_revision, preprocessing_version, input_hash, raw_output, raw_label, raw_score, normalized_label, normalized_score, calibrated_probability, calibration_version, quality, sufficiency, ood_status, status, reason_codes, started_at, completed_at
Lifecycle states (16)	planned, dispatched, running, completed, abstained, not_applicable, unavailable, timed_out, failed, cancelled, privacy_blocked, policy_skipped, insufficient, unsupported, degraded, budget_exceeded
Semantic rule	raw_score, normalized_score, and calibrated_probability are separate nullable fields and never silently substituted.

Receipt	Fields
inference_receipt.v2 fields	receipt_id, registry_resolution_id, deployment_id, deployment_bundle_id, request_id, provider, endpoint_identity, endpoint_deployment_revision, request_sha256, response_sha256, provider_request_id, components, attempt, latency_ms, billed_units, completion_status, received_at

Receipt	Fields
components[] fields	component_role, model_key, repository_id, immutable_revision, artifact_or_container_digest, preprocessing_version, input_schema_version, output_schema_version, execution_order, qualification_status
Compatibility	A valid v1 receipt is exposed as a v2 receipt with one synthetic component derived from its singular repository/model/revision/preprocessing fields; missing historical provenance remains explicit and is never fabricated.

11. HF failure engineering

Every attempt is deadline-bound and receipt-visible. Only transient transport/service failures may retry, and only against the same immutable deployment. Maximum attempts, exponential backoff, jitter, Retry-After handling, circuit-breaker thresholds and half-open probes are deployment-policy fields established by load/failure tests.

- Never create a retry storm or cross-provider failover.
- Cancellation propagates to queued work and provider attempts.
- Queue fallback is allowed only when request policy and freshness deadline permit.

Failure	Retry rule	Terminal behavior
401/403, 404, 409, 413	do not retry until credentials, deployment/config or payload changes	failed/unsupported evidence; alert owner
408, 429, 5xx, reset/DNS/TLS	bounded same-deployment retry with jitter, Retry-After and propagated deadline	circuit breaker; queue or review only by policy
cold start/initialization/quota	bounded wait/retry only inside role deadline; never alternate model/provider	async conversion, degraded/uncertain or review
malformed/partial/NaN/Inf/label	never retry as semantic recovery and never guess	schema rejection plus failed evidence/incident

12. Security, privacy and legal

- Scoped tokens and secret references; protected/private endpoints; egress allowlist.
- Payload minimization and no raw content in routine logs.
- Review provider retention, training use, subprocessors, residency, DPA and breach terms.
- Review model, dataset, custom-code and derivative licenses; unclear means REQUIRES LEGAL REVIEW.
- Prefer safe serialization; scan model/custom code; lock dependencies; SBOM and image digest.

13. Qualification, calibration and promotion

A provider response is not qualification. Golden contract tests prove I/O and labels. Reproducible benchmarks prove role performance and robustness. Held-out calibration proves probability semantics. Load/cold-start tests establish SLO and cost. Shadow/canary establish operational safety.

- No model-card metric is copied into the product as a VeriTrust result.
- Report failures/abstentions and confidence intervals.
- Promotion is signed; rollback is rehearsed.

14. Pricing, capacity and cost controls

Endpoint pricing is compute-based and minute-calculated from hourly instance rates. Serverless/provider pricing and credits vary. Actual VeriTrust costs remain UNKNOWN until account quota, region, instance, payload, traffic and cold-start tests exist. Monthly cost is computed from metered provider units plus endpoint instance-rate x active minutes/60, storage/egress and review labor; no fixed amount is asserted before those inputs exist.

Control	Policy
Serverless	request/unit budget, 429 handling, exact provider
Endpoint	instance/region, min/max replicas, scale-to-zero by role
Tenant	rate/concurrency and monthly budget
Receipt	billed units/cost estimate and reconciliation

Tier	Planning load	HF topology	Cost/shadow gate
T0 research	1 rpm / 1 concurrent (planning cap)	serverless where qualified; endpoints scale to zero	100% shadow permitted; cost UNKNOWN until test
T1 pilot	10 rpm / 3 concurrent (planning cap)	fast serverless; protected endpoints 0-1 min replica by role	25% shadow; measured invoice required
T2 small production	60 rpm / 10 concurrent (planning cap)	qualified fast + bounded endpoint pools	10% shadow; tenant/model budgets
T3 medium	600 rpm / 50 concurrent (planning envelope)	load-derived replica ranges and weighted queues	2-5% shadow; reserved budget/alerts
T4 enterprise	>600 rpm / 100+ concurrent (design input)	account/region/SLA-specific pools and isolation	<=1% shadow; contract and load test required

15. Acquisition and implementation checklist

- Re-query provider mapping and save timestamped result.
- Freeze exact source revision; review files, code, license and dataset.
- Choose serverless only if exact mapping/task/golden/privacy/quota gates pass.
- Otherwise create protected/private HF Endpoint with revision and handler/image digest.
- Register deployment before any request; validate receipts.
- Benchmark, calibrate, shadow, canary, promote and rehearse rollback.
- Never download weights into the VeriTrust production repository/runtime.

References and package evidence

Primary sources and immutable model snapshots are listed in Research/Sources/source_ledger.csv and Research/Sources/HuggingFace/. Repository claims are traceable through repository_inventory.csv, source_file_audit.csv, runtime_call_graph.md, current_architecture_map.md and code_claim_ledger.csv.

- Architecture_Source_of_Truth.yaml - canonical invariants, roles, schemas and rollout.
- Model_Registry_Proposal.json - immutable role/deployment proposal.
- HuggingFace_Deployment_Matrix.csv - point-in-time provider and endpoint decisions.
- Technical_Findings_Register.csv / Implementation_Backlog.csv - evidence-backed remediation.
- Database_Target_Contract.md / Database_Migration_Specification.md - logical contract without fabricated SQL.
- Competitor_Analysis.md / Competitor_Matrix.csv - primary-source market design loop.

Authoritative external references

Hugging Face Inference Providers | Hugging Face Inference Endpoints | C2PA Technical Specification. Access dates, exact page URLs, claim summaries and verification status are preserved in the source ledger.

Universal operational controls

Security, privacy, cost, capacity, calibration, observability, incident response and rollback are release gates for every role. No UNKNOWN is replaced with an estimate; unresolved values remain explicit until an evidence-producing test or owner decision closes them.

Final boundary

Hugging Face-managed learned inference is authoritative. No learned production model runs inside the VeriTrust Vercel application or a VeriTrust-owned worker/server; no production weights are required in the VeriTrust deployment repository.